

INTERNSHIP TASK REPORT

AUTOMATED DATA PIPELINE

Week-03 | Task 2 | Khansa Malik

1. Objective

The project implements a practical Python data pipeline for ingesting raw CSV and JSON data, validating records, cleaning accepted values, transforming selected fields, generating summary statistics, and exporting a clean dataset together with an error log.

2. Architecture

The solution follows a modular src package layout. `readers.py` handles CSV/JSON ingestion; `validators.py` applies field and record validation; `cleaners.py` normalizes validated data; `transformers.py` performs type/date/derived-field/currency transformations; `stats.py` computes summary statistics; `pipeline.py` orchestrates the complete workflow; and `cli.py` provides command-line access. Sample inputs are stored under `data/`, tests under `tests/`, and a programmatic example under `examples/`.

3. Quality and Testing

The project was designed for maintainability and verification rather than a single demonstration script. It includes shared pytest fixtures, unit tests for the major processing modules, and an end-to-end integration test. During development, 64 tests passing and approximately 88% overall coverage, with core processing modules at approximately 97-100% coverage. A defect involving duplicate counts in the summary was identified and followed by a code fix plus regression testing.

4. Expected Outputs

The pipeline is intended to produce a cleaned dataset, an error log for invalid records, and summary statistics for the resulting clean data. The included sample CSV and JSON inputs contain both valid and intentionally problematic records so that validation, error handling, and multi-format ingestion can be demonstrated.

5. Task Outcome

The implementation aligns with the practical requirements of Task 2 by separating pipeline concerns, supporting reusable execution, providing automated tests, and emphasizing reliable handling of invalid input. The repository is suitable for version control and GitHub upload after adding the standard repository files: `README.md`, `.gitignore`, and `LICENSE`.